THE INDEX OF SELF-DUAL TERNARY CODES

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Abstract. Gaiotto and Johnson-Freyd attached to every self-dual ternary code C a holomorphic $\mathcal{N}=1$ superconformal vertex operator algebra whose supersymmetric index is the number of even minus the number of odd codewords of full weight, $\mathrm{Index}(C)$. They conjectured that $\mathrm{Index}(C)$ is always divisible by 24 and that every length $12k$ carries a code of index 24, and they asked for a systematic construction of such codes. We prove both statements. Divisibility follows from a signed 2-design property of the full-weight codewords, which we obtain from a single Poisson summation. For existence we introduce a block-weighted index and a *partial quadrupling* of codes that preserves it; starting from the ternary Golay code it gives explicit codes of index 24 in every length $12k$. The construction commutes with code automorphisms that permute its blocks. Starting from the Golay code, from the length-36 code of Gaiotto and Johnson-Freyd, and from one code of length 72, we obtain for every k a code of index 24 whose automorphism group contains the groups for an orbifold construction of holomorphic $\mathcal{N}=1$ vertex operator superalgebras of the minimal index $24/\gcd(k,24)$: the cyclic groups of Gaiotto and Johnson-Freyd when $8\nmid k$, and the quaternion group Q_8, resp. $Q_8\times\mathbb{Z}/3$, when $8\mid k$. Cyclic groups cannot work when $8\mid k$: an automorphism g with $g^4=-1$ forces $48\mid\mathrm{Index}(C)$. The proofs of the two conjectured statements are formally verified in Lean.

1. Introduction

A ternary code of length n is a subspace $C\subseteq\mathbb{F}_3^n$. It is *self-dual* if $C=C^\perp$ for the standard form $\langle x,y\rangle=\sum_i x_iy_i$; self-dual ternary codes exist exactly when $4\mid n$. A codeword is *maximal* if none of its entries is 0, and it is *even* or *odd* according to the parity of the number of its entries equal to 1. Following Gaiotto and Johnson-Freyd [4, §1], the *index* of a self-dual ternary code is

$$\mathrm{Index}(C)=\#\{\text{even maximal codewords}\}-\#\{\text{odd maximal codewords}\}.$$

They show that C determines a holomorphic $\mathcal{N}=1$ vertex operator superalgebra $V_{\Lambda(C)}$ of central charge n, the lattice superalgebra of $\Lambda(C)=\{x/\sqrt{3}:x\in\mathbb{Z}^n,\ x\bmod 3\in C\}$, whose supersymmetric (Ramond–Ramond) index is $\mathrm{Index}(C)$ [4, Prop. 2.4]. It vanishes unless $12\mid n$. The Stolz–Teichner programme predicts that the index of a holomorphic $\mathcal{N}=1$ superconformal field theory of central charge $12k$ is divisible by $24/\gcd(k,24)$, the smallest multiple of Δ^k in the image of $\pi_{24k}\mathrm{TMF}\to\mathrm{MF}_{12k}$ [5, Prop. 4.6]. For codes this led to:

> **Conjecture** ([4, Conj. 1.4][1]). The index of any self-dual ternary code is divisible by 24, even if the length is merely divisible by 12. For all lengths $c=12k$, there exists a self-dual ternary code of index 24.

Gaiotto and Johnson-Freyd proved the first sentence when $24\mid n$, using Borcherds' theorem on theta series of even unimodular lattices; they deduced that $12\mid\mathrm{Index}(C)$ when $12\mid n$, and wrote that they did not know an elementary proof [4, Thm. 1.3]. With Elkies they found codes of index 24 for $k\le 5$ by computer search, and asked: *is there a systematic construction of self-dual ternary codes of index* 24 *which would avoid the use of intensive computation?* [4, Question 0.1]. We prove the conjecture and answer the question.

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[1]Numbering of statements in [4] refers to the arXiv version.

Theorem A. *For every self-dual ternary code C of positive length,* $24 \mid \mathrm{Index}(C)$.

Theorem B. *For every $k \geq 1$ there is an explicit self-dual ternary code of length $12k$ and index 24.*

The length must be positive: the code of length 0 has index 1, the empty word being an even maximal codeword.

Theorem A follows from a *signed 2-design* property of the maximal codewords: lifting them to $\varepsilon \in \{\pm 1\}^n$ with signs $s(\varepsilon) = \pm 1$ (even/odd), $\sum_\varepsilon s(\varepsilon)\varepsilon_i\varepsilon_j = \mathrm{Index}(C)\,\delta_{ij}$ (Theorem 3.1). This comes from one Poisson summation over C with two marked coordinates, and the factors 8 and 3 of 24 follow by short counting arguments. For Theorem B we attach to a code E and a set σ of disjoint 4-sets of coordinates a block-weighted index $R_\sigma(E)$, and we construct a self-dual code $Q_\sigma(E)$ of length $n + 12|\sigma|$ by gluing tetracodes onto the blocks with the map $(x, y) \mapsto (x + y, x - y)$, twice. Its index is $R_\sigma(E)$, and it carries a new block system with the same value of R (Lemma 4.6). Starting from the Golay code, this reaches every length $12k$.

Codes with symmetry. Gaiotto and Johnson-Freyd pose the *Main Question*: is there, for each k, a holomorphic $\mathcal{N} = 1$ vertex operator (super)algebra of central charge $12k$ and index $24/\gcd(k, 24)$? They answer it for $k \leq 5$ with the following theorem, where $\mathrm{Aut}(C)$ is the group of signed coordinate permutations preserving C; it contains -1.

> **Theorem** ([4, Thm. 5.6]). Let C be self-dual of length $12k$ with $\mathrm{Index}(C) = 24$, put $m = \gcd(24, k)$, and let $g \in \mathrm{Aut}(C)$ have order m. Suppose that (1) if m is even, $g^{m/2} = -1$, and (2) if $3 \mid m$, $g^{m/3}$ fixes as many even as odd maximal codewords and permutes the coordinates in 3-cycles only. Then g lifts to a non-anomalous automorphism of $V_{\Lambda(C)}$, and the orbifold $V_{\Lambda(C)}/\!\!/\langle g\rangle$ has index $24/m$.

The partial quadrupling commutes with code automorphisms that permute its blocks, and the *twined count* $T_h(C) = \sum_{x \in C,\, hx = x} f(x)$ of an automorphism h of order 3 (the signed count of h-fixed maximal codewords, f as in §2) is carried along as a block-weighted count on the h-fixed code (Lemma 6.5). With three seeds—the Golay code, the code of length 36 of [4, Ex. 5.7], and one code of length 72—we obtain:

Theorem C. *For every $k \geq 1$ there is an explicit self-dual ternary code C of length $12k$ with $\mathrm{Index}(C) = 24$ whose automorphism group contains*

(a) *if $3 \mid k$: an element h of order 3 without fixed coordinates and with $T_h(C) = 0$;*
(b) *if $4 \mid k$: elements J, K with $J^2 = K^2 = -1$ and $KJ = -JK$;*
(c) *if $12 \mid k$: elements h, J, K as in (a) and (b) with $hJ = Jh$ and $hK = Kh$.*

Corollary. *Let $m = \gcd(k, 24)$ with $8 \nmid k$, and let C be as in Theorem C. For $m = 1$, 2, 3, 4, 6, 12 the element $g = 1$, -1, h, J, $-h$, Jh of $\mathrm{Aut}(C)$ satisfies the hypotheses of* [4, Thm. 5.6]. *Hence $V_{\Lambda(C)}/\!\!/\langle g\rangle$ is a holomorphic $\mathcal{N} = 1$ vertex operator superalgebra of central charge $12k$ and index $24/\gcd(k, 24)$.*

This answers the Main Question for all k with $8 \nmid k$, within the orbifold framework of [4, §4] on which their Theorem 5.6 rests. For $8 \mid k$ the cyclic route is closed:

Theorem D. *Let C be a self-dual ternary code of length $12k$ and $g \in \mathrm{Aut}(C)$ with $g^4 = -1$. Then $48 \mid \mathrm{Index}(C)$, and $\mathrm{Index}(C) = 0$ if k is odd. In particular, if $8 \mid k$, no self-dual code of length $12k$ and index 24 has an automorphism g as in* [4, Thm. 5.6].

Indeed, for $m \in \{8, 24\}$ condition (1) gives $h = g^{m/8}$ with $h^4 = -1$. The quaternion group $Q_8 = \langle J, K\rangle$ of Theorem C(b) has no element of order 8, and every commuting pair in Q_8, or in $Q_8 \times \mathbb{Z}/3$, generates a cyclic group. If the actions of these groups on $V_{\Lambda(C)}$ are non-anomalous,

the orbifolds have index $24/8 = 3$ and $24/24 = 1$, the minimal values for $k \equiv 8, 16$ and $k \equiv 0 \pmod{24}$. Showing that they are non-anomalous for $8 \mid k$ is an anomaly computation for Q_8, carried out in a companion paper [2]; the present paper provides the codes.

Formal verification. Theorems A and B are formalized in Lean 4 with Mathlib [7, 8], as the single statement `GJF.conjecture_1_4` (§10), and so is Theorem D (`GJF.Obs.theoremD`). They use only the standard axioms; the index is defined literally as $\#\{\text{even}\} - \#\{\text{odd}\}$ maximal codewords.

Organisation. §2 collects notation. §3 proves Theorem A. §4 introduces partial quadrupling and §5 proves Theorem B. §6 makes the construction equivariant and §7 proves Theorem C. §8 proves Theorem D, §9 adds two congruences (including $R_\sigma(E) \equiv \mathrm{Index}(E) \pmod{48}$), and §10 describes the computations and the formalization. The seed data are in the appendix.

2. Preliminaries

2.1. **Codes and the index.** Let $\varphi\colon \mathbb{F}_3 \to \{0, \pm 1\}$ be $\varphi(0) = 0$, $\varphi(1) = -1$, $\varphi(2) = 1$, and $f(x) = \prod_i \varphi(x_i)$ for $x \in \mathbb{F}_3^n$. For $X \subseteq \mathbb{F}_3^n$ put $F(X) = \sum_{x \in X} f(x)$. The *weight* $\mathrm{wt}(x)$ is the number of nonzero entries.

Lemma 2.1. *Let $C \subseteq \mathbb{F}_3^n$ be self-dual, $n \geq 1$.*

(1) *$4 \mid n$, and every u in a self-orthogonal code has $\mathrm{wt}(u) \equiv 0 \pmod 3$.*
(2) *If C has a maximal codeword, then $12 \mid n$.*
(3) *$\mathrm{Index}(C) = F(C)$, and $f(-x) = f(x)$ for all $x \in \mathbb{F}_3^n$.*
(4) *If C' is self-dual as well, then $\mathrm{Index}(C \oplus C') = \mathrm{Index}(C)\,\mathrm{Index}(C')$.*

Proof. (1) C is a totally isotropic subspace of half dimension of the quadratic space $\mathbb{F}_3^n$, so $\mathbb{F}_3^n$ is a sum of $n/2$ hyperbolic planes, and comparing discriminants in $\mathbb{F}_3^\times/(\mathbb{F}_3^\times)^2$ gives $(-1)^{n/2} = 1$. Also $u \cdot u = \sum u_i^2 = \mathrm{wt}(u)$ in $\mathbb{F}_3$. (2) A maximal codeword has weight n. (3) $f(x) = (-1)^{\#\{i : x_i = 1\}}$ if x is maximal and 0 otherwise; $f(-x) = (-1)^n f(x) = f(x)$ as n is even. (4) f is multiplicative over direct sums. □

2.2. **Fourier analysis.** Let $\omega = e^{2\pi i/3}$ and $\sqrt{-3} = \omega - \omega^2$. For $h\colon \mathbb{F}_3^n \to \mathbb{C}$ put $\hat{h}(u) = \sum_x h(x)\omega^{u \cdot x}$. We use three facts:

(P) *Poisson summation:* for every subspace $D \subseteq \mathbb{F}_3^n$, $\sum_{x \in D^\perp} h(x) = |D|^{-1} \sum_{u \in D} \hat{h}(u)$.
(M) If $h(x) = \prod_i h_i(x_i)$, then $\hat{h}(u) = \prod_i \hat{h}_i(u_i)$.
(T) $\hat{\varphi} = \sqrt{-3} \cdot \varphi$, and $\iota = [\,\cdot \neq 0\,]$ has $\hat{\iota}(b) = 3[b = 0] - 1$. Hence $\hat{f} = (\sqrt{-3})^n f = 3^{n/2} f$ when $4 \mid n$.

(P) is $\sum_{u \in D} \omega^{u \cdot x} = |D|[x \in D^\perp]$; (T) is a one-line computation, with $(\sqrt{-3})^2 = -3$.

2.3. **Signed permutations.** $\mathrm{Mon}(n) = 2^n{:}S_n$ acts on $\mathbb{F}_3^n$ by $(gx)_{\pi(i)} = s_i x_i$ for $g = (\pi, s)$, $\pi \in S_n$, $s \in \{\pm 1\}^n$. We write $-1 = (\mathrm{id}, (-1, \ldots, -1))$, $\varepsilon(g) = \prod_i s_i$, and $\mathrm{Aut}(C) = \{g : gC = C\}$. Since $\varphi(-a) = -\varphi(a)$,

$$f(gx) = \varepsilon(g) f(x) \qquad (g \in \mathrm{Mon}(n),\ x \in \mathbb{F}_3^n). \tag{1}$$

Hence $\mathrm{Index}(dC) = -\mathrm{Index}(C)$ for a diagonal d with one entry -1; conjugation by d carries $\mathrm{Aut}(C)$ to $\mathrm{Aut}(dC)$, preserving relations, orders and cycle types, and $T_{dhd^{-1}}(dC) = -T_h(C)$. So a code of index ± 24 with some symmetry gives one of index $+24$ with the same symmetry. Also, if some $g \in \mathrm{Aut}(C)$ has $\varepsilon(g) = -1$, then $\mathrm{Index}(C) = 0$.

2.4. **The tetracode.** The *tetracode* $T \subseteq \mathbb{F}_3^4$ is spanned by $(1,1,1,0)$ and $(0,1,2,1)$. It is self-dual, its nonzero words have weight 3, and each 3-subset of the coordinates is the support of exactly two words. A *tetracode on* a 4-set Y is any self-dual code in $\mathbb{F}_3^Y$; these are the 8 monomial images of T and have the same properties.

3. The signed 2-design and the proof of Theorem A

For a maximal $x \in \mathbb{F}_3^n$ let $\varepsilon(x) \in \{\pm 1\}^n$ be given by $\varepsilon_k = -\varphi(x_k)$ (so $\varepsilon \equiv x \bmod 3$), and $s(\varepsilon) = f(x)$. Let $\mathcal{F}(C)$ be the set of these lifts of the maximal codewords. Then $\mathrm{Index}(C) = \sum_{\varepsilon\in\mathcal{F}} s(\varepsilon)$; $\mathcal{F} = -\mathcal{F}$ and $s(-\varepsilon) = s(\varepsilon)$ since n is even; and $\varepsilon\cdot\varepsilon' \equiv 0 \pmod 3$ for $\varepsilon, \varepsilon' \in \mathcal{F}$.

Theorem 3.1 (signed 2-design)**.** *For every self-dual ternary code C and all coordinates i, j,*

$$\sum_{\varepsilon\in\mathcal{F}(C)} s(\varepsilon)\,\varepsilon_i\varepsilon_j = \mathrm{Index}(C)\,\delta_{ij}.$$

Equivalently $\sum_{\varepsilon\in\mathcal{F}} s(\varepsilon)(\varepsilon\cdot\mu)^2 = \mathrm{Index}(C)|\mu|^2$ for all $\mu \in \mathbb{R}^n$.

The case $i = j$ is trivial. For $i \neq j$ put

$$g_{ij}(x) = \prod_{k\neq i,j} \varphi(x_k)\cdot[x_i \neq 0][x_j \neq 0], \qquad G_{ij}(X) = \sum_{x\in X} g_{ij}(x).$$

For maximal x we have $s(\varepsilon)\varepsilon_i\varepsilon_j = f(x)\varphi(x_i)\varphi(x_j) = g_{ij}(x)$, and g_{ij} vanishes on non-maximal words, so $G_{ij}(C) = \sum_{\mathcal{F}} s(\varepsilon)\varepsilon_i\varepsilon_j$. By (M) and (T),

$$\hat{g}_{ij}(u) = (\sqrt{-3})^{n-2} \prod_{k\neq i,j} \varphi(u_k)\cdot(3[u_i = 0] - 1)(3[u_j = 0] - 1). \tag{2}$$

Lemma 3.2. *If $3 \mid n$ and $D \subseteq \mathbb{F}_3^n$ is self-orthogonal, then $\hat{g}_{ij}(u) = (\sqrt{-3})^{n-2} g_{ij}(u)$ for every $u \in D$.*

Proof. If $u_k = 0$ for some $k \neq i, j$, both sides vanish. Otherwise $\mathrm{wt}(u) = n-2+[u_i \neq 0]+[u_j \neq 0]$, which is $\equiv 0 \pmod 3$ by Lemma 2.1; as $3 \mid n$ this forces $u_i \neq 0 \neq u_j$. Then the last two factors of (2) are $(-1)(-1) = 1$. □

Proof of Theorem 3.1. If $3 \nmid n$ there are no maximal codewords. Otherwise Poisson summation over $C = C^\perp$, $|C| = 3^{n/2}$, and Lemma 3.2 give

$$G_{ij}(C) = 3^{-n/2} \sum_{u\in C} \hat{g}_{ij}(u) = 3^{-n/2}(\sqrt{-3})^{n-2} G_{ij}(C) = \pm\tfrac{1}{3} G_{ij}(C),$$

so $G_{ij}(C) = 0$. □

Proposition 3.3. *Theorem 3.1 implies $8 \mid \mathrm{Index}(C)$ and $3 \mid \mathrm{Index}(C)$.*

Proof. Assume C has maximal codewords, so $12 \mid n$. *The factor* 8. For distinct a, b, c and $\varepsilon \in \{\pm 1\}^n$, $[\varepsilon_a = \varepsilon_b = \varepsilon_c] = \frac{1}{4}(1 + \varepsilon_a\varepsilon_b + \varepsilon_a\varepsilon_c + \varepsilon_b\varepsilon_c)$, so $\sum_{\varepsilon\in\mathcal{F},\ \varepsilon_a=\varepsilon_b=\varepsilon_c} s(\varepsilon) = \frac{1}{4}\mathrm{Index}(C)$. The summation set is stable under the fixed-point-free involution $\varepsilon \mapsto -\varepsilon$, which preserves s, so $\frac{1}{4}\mathrm{Index}(C)$ is even. *The factor* 3. Fix $\varepsilon^0 \in \mathcal{F}$. For $x \in \mathbb{Z}^n$ with $x \equiv \varepsilon^0 \pmod 3$ and $\varepsilon \in \mathcal{F}$, $\varepsilon\cdot x \equiv \varepsilon\cdot\varepsilon^0 \equiv 0 \pmod 3$, so $9 \mid \sum_{\mathcal{F}} s(\varepsilon)(\varepsilon\cdot x)^2 = \mathrm{Index}(C)|x|^2$. With $x = \varepsilon^0$ ($|x|^2 = n$) and $x = \varepsilon^0 + 3\varepsilon_1^0 e_1$ ($|x|^2 = n + 15$) we get $9 \mid 15\,\mathrm{Index}(C)$. □

Proof of Theorem A. Combine Theorem 3.1 and Proposition 3.3. □

Remark 3.4. (1) For the Golay code the 24 maximal codewords are all even (§5), and Theorem 3.1 says that the 12 of them taken up to sign are the rows of a Hadamard matrix of order 12 (cf. [1]). (2) The same computation with four marked coordinates $K = \{a,b,c,d\}$ gives $\sum_{\mathcal F} s(\varepsilon)\varepsilon_a\varepsilon_b\varepsilon_c\varepsilon_d = -\sum_u \prod_{k\notin K}\varphi(u_k)$, the sum over the codewords u that vanish at exactly three points of K and nowhere else; so the codewords of weight $n-3$ are exactly the obstruction to a signed 4-design (for the Golay code the left side is ± 8 for every K). (3) Theorem 3.1 can also be proved with modular forms: the theta series of the shadow of $\Lambda(C)$ with the harmonic coefficient v_iv_j, divided by η^n, is a holomorphic modular form of weight 2 for $\mathrm{SL}_2(\mathbb Z)$, hence 0; its leading coefficient is the left side of Theorem 3.1. That argument uses only the lattice, and gives $24 \mid \sum_v s(v)$ over the shadow vectors of norm k of every odd unimodular lattice of rank $12k$ whose shadow has minimum $\ge k$. (4) A proof of $8 \mid \mathrm{Index}(C)$ by Kneser neighbours is in Remark 9.3.

4. Partial quadrupling

4.1. **Two local identities.** For $a,b \in \mathbb F_3$ put $\psi(a,b) = \varphi(a+b)\varphi(a-b)$ and let $\kappa(a,b)$ be the number of nonzero entries of $(a+b, a-b)$. Then $\psi(a,b) \neq 0$ iff exactly one of a,b is nonzero, with $\psi(a,0) = 1$ and $\psi(0,b) = -1$ for $a,b \neq 0$; and $\kappa(a,b)$ is 2 if exactly one of a,b is nonzero, 1 if both are, and 0 if $a = b = 0$. Define $\lambda\colon \{0,\dots,4\} \to \mathbb Z$ by

$$\lambda(0) = 0, \quad \lambda(1) = -2, \quad \lambda(2) = \lambda(3) = 0, \quad \lambda(4) = 1.$$

Lemma 4.1 (doubling identity)**.** *For every tetracode T' on $\{1,2,3,4\}$ and $u \in \mathbb F_3^4$,*

$$\sum_{t\in T'} \prod_{i=1}^{4} \psi(u_i,t_i) = \lambda(\mathrm{wt}\, u).$$

Proof. A term is nonzero only if $\mathrm{supp}\, t$ is the complement of $\mathrm{supp}\, u$, and then it is $(-1)^{\mathrm{wt}\, t}$. Since T' has the weights 0 and 3, with two words on each 3-set, the sum is 1 if $\mathrm{wt}\, u = 4$, -2 if $\mathrm{wt}\, u = 1$, and 0 otherwise. □

Lemma 4.2 (refinement identity)**.** *Split $\{1,2,3,4\}$ into two pairs $\{p,q\}$, $\{r,s\}$. For every tetracode T' and $u \in \mathbb F_3^4$,*

$$\sum_{t\in T'} \lambda\big(\kappa(u_p,t_p)+\kappa(u_q,t_q)\big)\,\lambda\big(\kappa(u_r,t_r)+\kappa(u_s,t_s)\big) = \lambda(\mathrm{wt}\, u).$$

Proof. Let $L(t)$ be the summand. Since $\kappa(u_i,0) = 2[u_i \neq 0]$, $L(0) = [\mathrm{wt}\, u = 4]$. Let $t \neq 0$, so t vanishes at exactly one coordinate z; let z' be its partner and $\{x,y\}$ the other pair. On $\{x,y\}$ both κ's lie in $\{1,2\}$ and their sum is 4 iff $u_x = u_y = 0$. On $\{z,z'\}$ the sum is $2[u_z \neq 0] + \kappa(u_{z'},t_{z'})$; it is 4 iff $u_z \neq 0$, $u_{z'} = 0$, and 1 iff $u_z = 0$, $u_{z'} \neq 0$. Hence $L(t) = [u_x = u_y = 0]\big([u_z \neq 0, u_{z'} = 0] - 2[u_z = 0, u_{z'} \neq 0]\big)$, which vanishes unless $\mathrm{wt}\, u = 1$. If $\mathrm{supp}\, u = \{w\}$, the two words vanishing at w give $2\cdot 1$, the two vanishing at the partner of w give $2\cdot(-2)$, and the others 0: in total $-2 = \lambda(1)$. If $\mathrm{wt}\, u \neq 1$, only $t = 0$ contributes, and $[\mathrm{wt}\, u = 4] = \lambda(\mathrm{wt}\, u)$. □

4.2. **Block systems.**

Definition 4.3. A *block system* σ on a coordinate set X is a set of pairwise disjoint 4-subsets (*blocks*), each split into two pairs. The other coordinates are *singletons*. For $x \in \mathbb F_3^X$ and a code $E \subseteq \mathbb F_3^X$ put

$$w_\sigma(x) = \prod_{i \text{ singleton}} \varphi(x_i) \prod_{\beta\in\sigma} \lambda(\mathrm{wt}(x|_\beta)), \qquad R_\sigma(E) = \sum_{e\in E} w_\sigma(e).$$

The splittings do not enter R_σ, and $R_\emptyset(E) = F(E)$.

Definition 4.4 (partial quadrupling)**.** Let $E \subseteq \mathbb{F}_3^X$ be a code and σ a block system with blocks $\beta_1, \dots, \beta_b$.

(1) For each coordinate i of a block add a coordinate i^*, and choose a tetracode T_β on $\beta^* = \{i^* : i \in \beta\}$ for each block. Apply h_1, the map $(x_i, x_{i^*}) \mapsto (x_i + x_{i^*}, x_i - x_{i^*})$ on every pair $\{i, i^*\}$. For a block $\beta = \pi_1 \sqcup \pi_2$ the *step-one blocks* are $B_\pi = \pi \sqcup \pi^*$ for $\pi \in \{\pi_1, \pi_2\}$; there are $2b$ of them, and each is split into two pairs.
(2) For each coordinate y of a step-one block add a coordinate y^{**}, choose a tetracode T_B on B^{**} for each step-one block B, and apply h_2, the same map on every pair $\{y, y^{**}\}$.

Put $Q_\sigma(E) = h_2 h_1\big(E \oplus \bigoplus_\beta T_\beta \oplus \bigoplus_B T_B\big)$, a code of length $|X| + 12b$. Singletons are untouched. For a set $\mathcal{B}$ of step-one blocks let σ'' be the block system on the coordinates of $Q_\sigma(E)$ consisting, for each $B = \varpi_1 \sqcup \varpi_2 \in \mathcal{B}$ split into its pairs, of the two blocks $\varpi_1 \sqcup \varpi_1^{**}$ and $\varpi_2 \sqcup \varpi_2^{**}$ (split in any way).

The construction depends on the tetracodes and the splittings; by Lemmas 4.1 and 4.2 every choice works below.

Lemma 4.5. *If E is self-dual, so is $Q_\sigma(E)$.*

Proof. On one pair, $(x+y)(x'+y') + (x-y)(x'-y') = 2(xx' + yy') = -(xx' + yy')$, so h_1 and h_2 negate the form on the coordinates they pair. Write $v = (e, t_1, t_2)$ and $e = (e_S, e_B)$ (singleton and block parts). h_1 negates the form on (e_B, t_1), and h_2 negates it on (e_B, t_1, t_2): those are the coordinates of the step-one blocks and their partners. So $\langle h_2 h_1 v, h_2 h_1 v' \rangle = \langle e, e' \rangle + \langle t_1, t_1' \rangle - \langle t_2, t_2' \rangle = 0$, and the image has dimension $|X|/2 + 2b + 4b$, half the length. □

A single step would not suffice: h_1 negates the form on the blocks but not on the singletons.

Lemma 4.6 (main lemma)**.** *For every code E (self-dual or not), every block system σ, all choices in Definition 4.4, and every set $\mathcal{B}$ of step-one blocks,*

$$F(Q_\sigma(E)) = R_\sigma(E) \qquad \text{and} \qquad R_{\sigma''}(Q_\sigma(E)) = R_\sigma(E).$$

In particular $\operatorname{Index}(Q_\sigma(E)) = R_\sigma(E)$ *if E is self-dual.*

Proof. The first identity is the case $\mathcal{B} = \emptyset$ of the second. Write

$$R_{\sigma''}(Q_\sigma(E)) = \sum_{e, t_1, t_2} w_{\sigma''}\big(h_2 h_1(e, t_1, t_2)\big),$$

put $u = h_1(e, t_1)$ and sum over $t_2 = (t_B)_B$ first. The weight factors over the singletons of σ and over the step-one blocks B together with B^{**}, and t_B enters only the factor of B. If $B \notin \mathcal{B}$, its eight coordinates are singletons of σ'' and the factor is

$$\prod_{y \in B} \varphi(u_y + t_{B,y})\, \varphi(u_y - t_{B,y}) = \prod_{y \in B} \psi(u_y, t_{B,y});$$

summing over t_B gives $\lambda(\mathrm{wt}(u|_B))$ by Lemma 4.1. If $B \in \mathcal{B}$, the factor is

$$\lambda\Big(\sum_{y \in \varpi_1} \kappa(u_y, t_{B,y})\Big)\, \lambda\Big(\sum_{y \in \varpi_2} \kappa(u_y, t_{B,y})\Big),$$

since the weight of h_2 on a pair (y, y^{**}) is $\kappa(u_y, t_{B,y})$; summing over t_B gives $\lambda(\mathrm{wt}(u|_B))$ by Lemma 4.2. Now sum over $t_1 = (t_\beta)_\beta$. For $\beta = \pi_1 \sqcup \pi_2$, $\mathrm{wt}(u|_{B_\pi}) = \sum_{i \in \pi} \kappa(e_i, t_{\beta, i^*})$, so the two step-one blocks of β give the summand of Lemma 4.2, and summing over t_β gives $\lambda(\mathrm{wt}(e|_\beta))$. What remains is $\sum_{e \in E} w_\sigma(e) = R_\sigma(E)$. □

Remark 4.7 (the gadget)**.** Per block, Q_σ glues in a fixed code of length 16. Let $\Gamma = h_2h_1(0 \oplus T_\beta \oplus T_{B_1} \oplus T_{B_2})$, a self-orthogonal code of dimension 6 on the 16 coordinates arising from β. The map $u \mapsto h_2h_1(u, 0, 0) + \Gamma$ is an isometry of $\mathbb{F}_3^4$ onto $\Gamma^\perp/\Gamma$ (two anti-isometries compose to an isometry), and the proof of Lemma 4.6 shows that the coset of u has signed full-weight count $\lambda(\operatorname{wt} u)$. So $Q_\sigma(E)$ is the gluing of E with $|\sigma|$ copies of Γ, and "+12 per block" is $16 - 4$.

5. Index 24 in every length

Let $G_{12} \subseteq \mathbb{F}_3^{12}$ be the extended ternary Golay code with generator matrix

$$\begin{pmatrix} 1&0&0&0&0&0&0&1&1&1&1&1\\ 0&1&0&0&0&0&2&0&1&2&2&1\\ 0&0&1&0&0&0&2&1&0&1&2&2\\ 0&0&0&1&0&0&2&2&1&0&1&2\\ 0&0&0&0&1&0&2&2&2&1&0&1\\ 0&0&0&0&0&1&2&1&2&2&1&0 \end{pmatrix}$$

[4, Ex. 1.5], coordinates $0, \dots, 11$. Its weight enumerator is $1 + 264y^6 + 440y^9 + 24y^{12}$, and the supports of the weight-6 words are the 132 hexads of a Steiner system $S(5, 6, 12)$ [3, Ch. 10, 11]. As the minimum weight is 6, two words of weight 6 (resp. 9) with the same support are equal or opposite; so each hexad carries exactly two words $\pm c$, and, since $440 = 2\binom{12}{3}$, each 3-set is the zero set of exactly two words of weight 9. All 24 maximal codewords are even, so $\operatorname{Index}(G_{12}) = 24$.

Proposition 5.1. *Let σ_1 consist of the block $\{0, 1, 2, 3\}$ and σ_2 of the blocks $\{0, 1, 2, 3\}$ and $\{4, 5, 6, 8\}$. Then $R_{\sigma_1}(G_{12}) = R_{\sigma_2}(G_{12}) = 24$.*

Proof. Direct evaluation over the 729 codewords. (More generally $R_\sigma(G_{12}) = \pm 24$ for all 495 single blocks, and for 13,365 of the 17,325 two-block systems; the others give ± 72.) □

Lemma 5.2 (engine)**.** *If E is self-dual of length m and σ is a two-block system with $R_\sigma(E) = \pm 24$, then $E' = Q_\sigma(E)$ is self-dual of length $m + 24$ with $\operatorname{Index}(E') = \pm 24$, and it carries a two-block system σ' with $R_{\sigma'}(E') = R_\sigma(E)$.*

Proof. Lemmas 4.5 and 4.6, with $\mathcal{B}$ a single step-one block. □

Proof of Theorem B. $k = 1$: G_{12}. Odd $k = 2j + 1 \geq 3$: apply Lemma 5.2 j times to (G_{12}, σ_2). Even $k = 2j + 2$: $Q_{\sigma_1}(G_{12})$ has length 24 and index 24, and refining one of its two step-one blocks gives a two-block system with $R = 24$ (Lemma 4.6); apply Lemma 5.2 j times. Finally re-sign one coordinate if the index is -24. □

Remark 5.3. (1) The codes have minimum distance at least 6: if w is a codeword of weight 3 and x is maximal, then $\langle x, w\rangle = 0$ forces x to agree with w or with $-w$ on $\operatorname{supp} w$, so one of $x \pm w$ is maximal of the opposite parity, and w pairs off the maximal codewords. (2) The codes are built from G_{12} and tetracodes only, and a generator matrix is computable in time polynomial in k; no search is needed beyond the 729-word check of Proposition 5.1. Lengths 72 and 84 are covered neither by the examples of [4] ($k \leq 5$) nor by doubling ($k = 2^j$, the case σ without singletons). (3) Through [4] the codes give holomorphic $\mathcal{N} = 1$ vertex operator superalgebras of index 24 in every central charge $12k$.

For partitions of the Golay coordinates into three blocks the value of R has a simple description, which we use in §7.

Lemma 5.4. *Let σ be a partition of $\{0, \dots, 11\}$ into three blocks, and let $H(\sigma)$ be the number of hexads that contain one block of σ and meet each of the other two in one point. Then $R_\sigma(G_{12}) = 8H(\sigma) - 24$.*

Proof. There are no singletons, and $\prod_\beta \lambda(|\operatorname{supp} c \cap \beta|) \neq 0$ requires every intersection to have 1 or 4 points. The 24 words of weight 12 contribute 1 each. A word of weight 9 contributes iff its zero set lies in one block, and then $1 \cdot 1 \cdot (-2)$; there are $3 \cdot 4$ such zero sets, each carrying two words, in total -48. A word of weight 6 contributes iff its hexad contains one block and meets the others once, and then $1 \cdot (-2)^2 = 4$; with two words per hexad this is $8H(\sigma)$. □

Over all 5775 partitions, $R_\sigma(G_{12})$ takes the values -24, 24, 72 on 495, 3960 and 1320 of them.

6. Codes with symmetry

Throughout, $E \subseteq \mathbb{F}_3^X$ is a code on a finite set X, σ a block system on X, and $\mathrm{Mon}(X)$ the group of signed permutations of $\mathbb{F}_3^X$. For a subset Y preserved by g, $g|_Y$ is the restriction.

6.1. **Compatible groups.**

Definition 6.1. A finite group $G \subseteq \mathrm{Aut}(E)$ with $-1 \in G$ is *σ-compatible* if every $g \in G$ maps each block of σ onto a block, and each pair of a block onto a pair of the image block. For $\beta \in \sigma$ let G_β be its stabiliser and $\rho_\beta(G_\beta) = \{g|_\beta\} \subseteq \mathrm{Mon}(\beta)$. The block β has *type*

- τ_0 if $\rho_\beta(G_\beta) = \{\pm 1\}$;
- τ_J if $\rho_\beta(G_\beta) = \{\pm 1, \pm j\}$ with $j^2 = -1$ and j mapping each pair of β onto itself;
- τ_Q if $\rho_\beta(G_\beta) = \{\pm 1, \pm j, \pm k, \pm jk\}$ with $j^2 = k^2 = -1$, $kj = -jk$, and the pairs of β are the two orbits of $\langle j \rangle$ on β.

In type τ_J, j swaps the two points of each pair (a fixed point p would give $j^2 e_p = e_p$). In type τ_Q, j, k and jk induce the three fixed-point-free involutions of β, and $\rho_\beta(G_\beta)$ permutes the eight vectors $\pm e_p$ ($p \in \beta$) simply transitively.

Lemma 6.2 (invariant tetracodes)**.** *Let Y be a 4-set split into two pairs and $H \subseteq \mathrm{Mon}(Y)$ a group as in one of the three types. Then some tetracode on Y is H-invariant.*

Proof. Type τ_0: every code is invariant under ± 1. Type τ_J: write the pairs as $\{p, q\}$, $\{r, s\}$ with $j(e_p) = a e_q$ and $j(e_r) = b e_s$; then $j(e_q) = -a e_p$ and $j(e_s) = -b e_r$. After the diagonal sign change $e_q \mapsto a e_q$, $e_s \mapsto b e_s$, which permutes the tetracodes on Y, we may take $a = b = 1$, so $j(x_p, x_q, x_r, x_s) = (-x_q, x_p, -x_s, x_r)$. It maps $(1, 1, 1, 0)$ and $(0, 1, 2, 2)$ to $(2, 1, 0, 1)$ and $(2, 0, 1, 2)$, which lie in their span; so that tetracode is j-invariant. Type τ_Q: let $A = \mathbb{F}_3\langle i, j \rangle$ be the quaternion algebra with basis $1, i, j, k = ij$ ($i^2 = j^2 = -1$, $ji = -ij$) and $Q = \{\pm 1, \pm i, \pm j, \pm k\}$. Fix $p \in Y$. By simple transitivity there is an isomorphism $\theta \colon Q \to H$, and $\Theta(x) = \theta(x) e_p$ for $x \in Q$ extends to a linear bijection $\Theta \colon A \to \mathbb{F}_3^Y$ with $\Theta(ya) = \theta(y)\Theta(a)$ ($y \in Q$, $a \in A$). So the H-invariant subspaces of $\mathbb{F}_3^Y$ are the images of the left ideals of A. As Θ maps $1, i, j, k$ to $\pm$ the standard basis, the standard form becomes (twice) the polar form of the reduced norm $a^2 + b^2 + c^2 + d^2$. Under $A \cong M_2(\mathbb{F}_3)$ (e.g. $i \mapsto \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $j \mapsto \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$) the reduced norm is the determinant. The left ideal of matrices whose rows lie in a fixed line of $\mathbb{F}_3^2$ is 2-dimensional and consists of singular matrices, so its image is a totally isotropic code of dimension 2 in $\mathbb{F}_3^Y$, i.e. an H-invariant tetracode. □

6.2. **Equivariant quadrupling.**

Construction 6.3. Let G be σ-compatible, with all blocks of types τ_0, τ_J, τ_Q. Make the choices of Definition 4.4 as follows.

(i) *Extension.* For $g = (\pi, s) \in G$ let $g^\sharp$ act on the enlarged coordinate set by g on X and by letting partners follow with the same sign: $g^\sharp(e_{i^*}) = s_i e_{\pi(i)^*}$ and $g^\sharp(e_{y^{**}}) = s'_y e_{\pi'(y)^{**}}$, where (π', s') is $g^\sharp$ on the coordinates of step one.

(ii) *Splittings.* The step-one block $B_\pi = \pi \sqcup \pi^*$ is split into the pairs π and π^*.

(iii) *Tetracodes.* On one block β of each G-orbit choose a tetracode T_β on β^* invariant under $\{g^\sharp|_{\beta^*} : g \in G_\beta\}$ (a copy of $\rho_\beta(G_\beta)$), and put $T_{g\beta} = g^\sharp T_\beta$. Likewise for step-one blocks, with the stabiliser G_B acting on B^{**}.

(iv) *Refinement.* $\mathcal{B}$ is a $G^\sharp$-stable set of step-one blocks. By (ii) the blocks of σ'' are $\pi \sqcup \pi^{**}$ and $\pi^* \sqcup \pi^{***}$ for $B_\pi \in \mathcal{B}$ (where $\pi^{***} = (\pi^*)^{**}$), and we split them into the pairs $\{\pi, \pi^{**}\}$ and $\{\pi^*, \pi^{***}\}$.

Lemma 6.4 (equivariant quadrupling)**.** *In Construction 6.3:*

(1) *the tetracodes in (iii) exist and are well defined;*
(2) *$g \mapsto g^\sharp$ is an injective homomorphism $G \to \mathrm{Aut}(Q_\sigma(E))$ with $(-1)^\sharp = -1$, and $g^\sharp$ has a fixed coordinate iff g has one;*
(3) *σ'' is $G^\sharp$-compatible, and its blocks have type τ_0 or τ_J.*

In particular every relation among elements of G holds among their extensions.

Proof. The extension is multiplicative because partners follow partners with equal signs, and injective because $g^\sharp|_X = g$; a new coordinate is fixed iff its partner is. $g^\sharp$ preserves X, the starred and the double-starred coordinates, so the stabiliser of B_π is the stabiliser G_π of the pair π. Let π be a pair of a block β. Then $\rho_\pi(G_\pi) \subseteq \{\pm 1, \pm j_\pi\}$ for a swap j_π of π with $j_\pi^2 = -1$: this is clear for types τ_0 and τ_J, and in type τ_Q the elements of $\rho_\beta(G_\beta)$ preserving the $\langle j\rangle$-orbit π are $\pm 1, \pm j$. Hence G_π acts on $B_\pi = \pi \sqcup \pi^*$ through $m \oplus m^*$ with $m \in \{\pm 1, \pm j_\pi\}$, of type τ_0 or τ_J for the pairs π, π^*. By Lemma 6.2 the tetracodes in (iii) exist, and $T_{g\beta}$ does not depend on g because T_β is G_β-invariant; so $E \oplus \bigoplus T_\beta \oplus \bigoplus T_B$ is $G^\sharp$-invariant. On a pair (i, i^*) the element $g^\sharp$ acts by the common sign s_i and maps it to the pair $(\pi(i), \pi(i)^*)$; since $(x, y) \mapsto (x + y, x - y)$ commutes with a common sign, h_1 and h_2 commute with $G^\sharp$, and $Q_\sigma(E)$ is $G^\sharp$-invariant. For (3): the stabiliser of $\pi \sqcup \pi^{**}$ is G_π, acting through $m \oplus m^{**}$, of type τ_0 or τ_J for the pairs π, π^{**}; similarly for $\pi^* \sqcup \pi^{***}$. □

The choice (ii) of splittings matters. With the splitting $\{p, p^*\} \sqcup \{q, q^*\}$ of B_π the new blocks would have smaller stabilisers, the G-orbits of step-one blocks would double in size from the next step on, and the lengths reached in §7 would skip residue classes. And a stabiliser acting on a pair by an involutive swap or by $\mathrm{diag}(1, -1)$ would admit no invariant tetracode at step two; the three types exclude this.

6.3. **Twined counts.** Let $h \in G$ have order 3 and no fixed coordinate. A permutation of order 3 of a 4-set has a fixed point, so h moves every block and the $\langle h\rangle$-orbits on σ have three elements. For x fixed by h, $\mathrm{wt}(x|_\beta)$ is constant on each orbit. Put

$$R^h_\sigma(E) = \sum_{x \in E,\ hx = x} \prod_{i \text{ singleton}} \varphi(x_i) \prod_{O \in \sigma/\langle h\rangle} \lambda\big(\mathrm{wt}(x|_{\beta_O})\big),$$

with one factor per orbit O ($\beta_O \in O$ arbitrary). Then $R^h_\emptyset(E) = T_h(E)$.

Lemma 6.5 (twined counts)**.** *In Construction 6.3 let $h \in G$ have order 3 and no fixed coordinate. Then*

$$T_{h^\sharp}(Q_\sigma(E)) = R^h_\sigma(E) \qquad \text{and} \qquad R^{h^\sharp}_{\sigma''}(Q_\sigma(E)) = R^h_\sigma(E).$$

Proof. Let $V = E \oplus \bigoplus T_\beta \oplus \bigoplus T_B$ and $\mathsf{H} = h_2 h_1$, so $Q_\sigma(E)^{h^\sharp} = \mathsf{H}(V^{h^\sharp})$. For an orbit $\{\beta, h\beta, h^2\beta\}$ the fixed vectors of $T_\beta \oplus T_{h\beta} \oplus T_{h^2\beta}$ are $(t, h^\sharp t, (h^\sharp)^2 t)$ with $t \in T_\beta$, and likewise for step-one blocks. For $\beta \in \sigma$ let $\hat\beta$ be the 16 coordinates arising from β, and for $y \in Q_\sigma(E)$ let $W_\beta(y)$ be the product of the σ''-weights (φ on singletons, λ on blocks) over $\hat\beta$. For each $i \in \beta$ the four coordinates i, i^*, i^{**}, i^{***} are either all singletons of σ'' or all in blocks, and $h^\sharp$ has the same sign on all four. So for $h^\sharp$-fixed y the φ-parts of $W_\beta, W_{h\beta}, W_{h^2\beta}$ coincide, and the λ-parts depend

only on supports. The weight in $R^{h\sharp}_{\sigma''}$ (φ on all singletons, one λ per orbit of blocks) therefore gives each orbit of σ the factor (φ-part)$^3\cdot$(λ-part) $= W_\beta(y)$, as the φ-part lies in $\{0,\pm1\}$. Hence

$$R^{h\sharp}_{\sigma''}(Q_\sigma(E)) = \sum_{x\in E^h}\ \prod_{i\ \text{singleton}}\varphi(x_i)\prod_O\ \sum_t W_{\beta_O}\big(\mathsf{H}(x,t)\big),$$

where t runs over the tetracode variables of β_O and its two step-one blocks. The inner sum is the local sum computed in the proof of Lemma 4.6: it equals $\lambda(\mathrm{wt}(x|_{\beta_O}))$. The first identity is the case $\mathcal{B}=\emptyset$. □

6.4. **Chains.**

Proposition 6.6. *Let E_0 be self-dual of length n_0, and G a σ_0-compatible group whose blocks have types τ_0,τ_J,τ_Q. Put $E_1 = Q_{\sigma_0}(E_0)$ (Construction 6.3) and, for $j\ge1$, let σ_j be the refinement of the G-orbit $\mathcal{B}_j$ of one step-one block of the construction of E_j, and $E_{j+1}=Q_{\sigma_j}(E_j)$. Then for all $j\ge1$:*

1. *E_j is self-dual of length n_j, with $n_1 = n_0+12|\sigma_0|$ and $n_{j+1}=n_j+24|\mathcal{B}_j|$;*
2. *G acts faithfully on E_j by the iterated extensions, with the same relations, and an element has a fixed coordinate on E_j iff it has one on E_0;*
3. *$\mathrm{Index}(E_j) = R_{\sigma_0}(E_0)$;*
4. *$T_h(E_j) = R^h_{\sigma_0}(E_0)$ for every $h\in G$ of order 3 without fixed coordinates.*

Proof. Induction on j, with Lemmas 4.5, 4.6, 6.4 and 6.5: by Lemma 4.6, $\mathrm{Index}(E_{j+1}) = R_{\sigma_j}(E_j)$ and $R_{\sigma_j}(E_j) = R_{\sigma_{j-1}}(E_{j-1})$, and likewise for the twined counts. For $j\ge1$ the blocks of σ_j have types τ_0 or τ_J, so Construction 6.3 applies again. □

7. Three families, and the proof of Theorem C

7.1. **The quaternion group on the Golay code.** Let $J,K\in\mathrm{Mon}(12)$ be given by $J(e_i) = s_ie_{J(i)}$, $K(e_i)=t_ie_{K(i)}$ with

$$J=(0\,1)(2\,3)(4\,5)(6\,7)(8\,9)(10\,11),\quad s=(-,+,-,+,+,-,-,+,+,-,-,+),$$
$$K=(0\,2)(1\,3)(4\,6)(5\,7)(8\,11)(9\,10),\quad t=(-,+,+,-,-,-,+,+,+,-,+,-).$$

Then $J,K\in\mathrm{Aut}(G_{12})$, $J^2=K^2=-1$ and $KJ=-JK$; so $\langle J,K\rangle\cong Q_8$, and it acts on the coordinates through a fixed-point-free Klein group with the orbits $\{0,1,2,3\}$, $\{4,5,6,7\}$, $\{8,9,10,11\}$. (There are 1320 such subgroups of $\mathrm{Aut}(G_{12})\cong2.M_{12}$, all conjugate.)

The obvious block system does not work:

Proposition 7.1. *For every $Q_8=\langle J,K\rangle\subseteq\mathrm{Aut}(G_{12})$ with $J^2=K^2=-1$, the partition σ into its three coordinate orbits has $R_\sigma(G_{12})=72$.*

Proof. Every 4-set lies in exactly four hexads, so by Lemma 5.4 it suffices to show that each hexad through an orbit O meets the other two orbits once. If a hexad $X\supseteq O$ met another orbit in two points x,y, the involution t of the Klein group with $t(x)=y$ would preserve O and $\{x,y\}$, hence X. A lift $\tilde t\in Q_8$ of t would map the word c supported on X to $\pm c$, and $\tilde t^2=-1$ would give $c=-c$. □

A mixed system does work. Let

$$\sigma_Q=\big\{\{0,1\}\sqcup\{2,3\},\ \{4,5\}\sqcup\{10,11\},\ \{6,7\}\sqcup\{8,9\}\big\}.$$

The first block is an orbit, of type τ_Q; the other two consist of one J-pair from each of the remaining orbits, have stabiliser $\langle J\rangle$ and type τ_J, and are swapped by K. So Q_8 is σ_Q-compatible, and $R_{\sigma_Q}(G_{12})=-24$ (direct evaluation, or $H(\sigma_Q)=0$ in Lemma 5.4).

Proposition 7.2. *For every $k \equiv 0 \pmod 4$ there is a self-dual code $\mathcal{Q}_k$ of length $12k$ with $\mathrm{Index}(\mathcal{Q}_k) = 24$ and $J, K \in \mathrm{Aut}(\mathcal{Q}_k)$ with $J^2 = K^2 = -1$, $KJ = -JK$.*

Proof. Apply Proposition 6.6 to (G_{12}, σ_Q, Q_8). The step-one blocks have stabiliser $\langle J\rangle$ of order 4, so $|\mathcal{B}_j| = 2$; the lengths are $n_1 = 12 + 36 = 48$ and $n_{j+1} = n_j + 48$, i.e. $n_j = 48j$. The index is $R_{\sigma_Q}(G_{12}) = -24$; re-sign one coordinate (§2). □

7.2. **The length-36 code of Gaiotto and Johnson-Freyd with $\mathbb{Z}/6$.** Let E_{36} be the self-dual code of length 36 of [4, Ex. 5.7(3)] (Appendix A.1), and h the coordinate permutation $i \mapsto i + 12 \pmod{36}$. Then $h \in \mathrm{Aut}(E_{36})$ has order 3 and no fixed coordinate, $\mathrm{Index}(E_{36}) = 24$ and $T_h(E_{36}) = 0$ ([4]; we have rechecked these facts), and $G = \langle -1, h\rangle \cong \mathbb{Z}/6$. Every block has an h-orbit of size 3, so all blocks have type τ_0, and any splittings, transported along h, make G compatible. Put

$$\beta = \{0, 2, 5, 25\}, \qquad \beta_1 = \{2, 15, 21, 28\}, \qquad \beta_2 = \{5, 6, 10, 31\},$$

$\sigma_A = \{\beta, h\beta, h^2\beta\}$ and $\sigma_B = \{\beta_1, h\beta_1, h^2\beta_1, \beta_2, h\beta_2, h^2\beta_2\}$.

Proposition 7.3. *$R_{\sigma_A}(E_{36}) = R_{\sigma_B}(E_{36}) = 24$ and $R^h_{\sigma_A}(E_{36}) = R^h_{\sigma_B}(E_{36}) = 0$. Consequently for every $k \equiv 0 \pmod 3$ there is a self-dual code $\mathcal{Z}_k$ of length $12k$ with $\mathrm{Index}(\mathcal{Z}_k) = 24$, and $h \in \mathrm{Aut}(\mathcal{Z}_k)$ of order 3 without fixed coordinates and with $T_h(\mathcal{Z}_k) = 0$.*

Proof. The four values are finite computations (§10). Put $\mathcal{Z}_3 = E_{36}$ and apply Proposition 6.6 to (E_{36}, σ_A, G) and (E_{36}, σ_B, G). The step-one blocks have stabiliser $\{\pm 1\}$, so $|\mathcal{B}_j| = 3$. From σ_A: $n_1 = 72$ and $n_{j+1} = n_j + 72$, giving all $k \equiv 0 \pmod 6$. From σ_B: $n_1 = 108$ and then $+72$, giving all odd $k \equiv 0 \pmod 3$, $k \geq 9$. □

Among the 13,365 h-stable systems of three blocks, 803 have $R = \pm 24$ and $R^h = 0$, so good choices are plentiful.

7.3. **A code of length 72 with $Q_8 \times \mathbb{Z}/3$.** Let E_{72} be the code of Appendix A.2: $72 = 18 \cdot 4$ coordinates in blocks $\{4b, \ldots, 4b + 3\}$, J, K acting on every block by left multiplication with the quaternions i, j, and h permuting the blocks by $3s + r \mapsto 3s + (r + 1 \bmod 3)$. Then E_{72} is self-dual, $J, K, h \in \mathrm{Aut}(E_{72})$, $J^2 = K^2 = -1$, $KJ = -JK$, $h^3 = 1$, h commutes with J, K and has no fixed coordinate; so $G = \langle J, K, h\rangle \cong Q_8 \times \mathbb{Z}/3$. Let σ_{QZ} consist of the blocks $\{4b, 4b+1\} \sqcup \{4b+2, 4b+3\}$, $0 \leq b \leq 5$ (two h-orbits). They are orbits of Q_8 on the coordinates, split into $\langle J\rangle$-orbits, so they have type τ_Q.

Proposition 7.4. *$R_{\sigma_{QZ}}(E_{72}) = 24$ and $R^h_{\sigma_{QZ}}(E_{72}) = 0$. Consequently for every $k \equiv 0 \pmod{12}$ there is a self-dual code $\mathcal{QZ}_k$ of length $12k$ with $\mathrm{Index}(\mathcal{QZ}_k) = 24$ and $J, K, h \in \mathrm{Aut}(\mathcal{QZ}_k)$ with the relations above.*

Proof. The two values are computations (§10). In Proposition 6.6 the step-one blocks have stabiliser $\langle J\rangle$ of order 4, so $|\mathcal{B}_j| = 6$, $n_1 = 144$ and $n_{j+1} = n_j + 144$. □

7.4. **Proof of Theorem C.** If $12 \mid k$ take $\mathcal{QZ}_k$. If $4 \mid k$ and $3 \nmid k$ take $\mathcal{Q}_k$. If $3 \mid k$ and $4 \nmid k$ take $\mathcal{Z}_k$. Otherwise take a code from Theorem B. □

Proof of the Corollary. $m = 1$ and $m = 2$ are immediate ($-1 \in \mathrm{Aut}(C)$ for every code). $m = 3$: $g = h$. $m = 4$: $g = J$ has order 4 and $g^2 = -1$. $m = 6$: $g = -h$ has order 6, $g^3 = -1$, and $g^2 = h^2$ permutes the coordinates in 3-cycles with $T_{h^2} = T_h = 0$. $m = 12$: $g = Jh$ has order 12, $g^6 = J^2 = -1$ and $g^4 = h$. In each case $\mathrm{Index}(C) = 24$ and the hypotheses of [4, Thm. 5.6] hold. □

8. Automorphisms with $g^4 = -1$: proof of Theorem D

Let $g \in \mathrm{Aut}(C)$ with $g^4 = -1$, $n = 12k$. On a cycle of length ℓ with sign product ϵ, g^ℓ acts as ϵ; so $g^4 = -1$ forces every cycle to have length 4 and $\epsilon = -1$. There are $3k$ cycles, so $\varepsilon(g) = (-1)^{3k}$, and by (1) $\mathrm{Index}(C) = 0$ if k is odd. Let k be even. Then $f \circ g = f$, and $\langle g \rangle \cong \mathbb{Z}/8$ acts freely on $\mathbb{F}_3^n \setminus \{0\}$ (every nontrivial subgroup contains $g^4 = -1$), so $F(X) \in 8\mathbb{Z}$ for every g-invariant $X \subseteq \mathbb{F}_3^n$.

Structure. Over $\mathbb{F}_3$, $x^4 + 1 = (x^2 + x + 2)(x^2 + 2x + 2)$, and on each 4-cycle the characteristic polynomial of g is $x^4 + 1$. So $\mathbb{F}_3^n = V_+ \oplus V_-$ with $V_\pm$ the kernels of the two factors at g, each the sum over the cycles of 2-dimensional pieces. Since g is an isometry and the roots of $x^2 + x + 2$ are inverse to those of $x^2 + 2x + 2$, both $V_\pm$ are totally isotropic, and they are dually paired. Hence the g-invariant self-dual codes are $C = C_+ \oplus C_-$ with $C_+ \subseteq V_+$ any g-invariant subspace and C_- its annihilator in V_-. On V_+, g acts as a scalar $\lambda \in \mathbb{F}_9$ of order 8, so C_+ is any $\mathbb{F}_9$-subspace. The element $J = g^2$ acts on $\mathbb{F}_3^n$ as multiplication by $\lambda^2 = \sqrt{-1}$; it is an isometry with $J^2 = -1$, so $\mathbb{F}_3^n$ is an $\mathbb{F}_9$-vector space and the form is the trace of a Hermitian form.

A four-code identity. Let $D \subseteq C$ be J-invariant of codimension 2. Then $D^\perp/D$ is a nondegenerate Hermitian plane over $\mathbb{F}_9$. Its ternary form is hyperbolic, with 32 nonzero isotropic vectors, and these form exactly the 4 isotropic $\mathbb{F}_9$-lines. So the J-invariant self-dual codes $C_1, \dots, C_4 \supseteq D$ contain all isotropic vectors of $D^\perp$, each outside D in exactly one of them. Maximal vectors are isotropic, and Poisson summation with $\hat{f} = 3^{n/2} f$ gives $F(D^\perp) = 9F(D)$. Hence

$$\mathrm{Index}(C_1) + \mathrm{Index}(C_2) + \mathrm{Index}(C_3) + \mathrm{Index}(C_4) = 12F(D). \tag{3}$$

Induction. If $C_+ \neq 0$, let $D_+ \subset C_+$ be an $\mathbb{F}_9$-hyperplane and $D = D_+ \oplus C_-$. Then $D^\perp/D = L_+ \oplus L_-$ with $L_\pm$ the images of the g-eigenlines, on which g acts by λ and $-\lambda$. The four codes through D are $C_1 = C$, $C_2 = D_+ \oplus D_+^\circ$ (D_+° the annihilator of D_+ in V_-), and two codes C_3, C_4 whose lines $\mathbb{F}_9(u + au')$ ($u \in L_+$, $u' \in L_-$, $a \neq 0$) are exchanged by g. As $f \circ g = f$, $\mathrm{Index}(C_3) = \mathrm{Index}(C_4)$, which lies in $24\mathbb{Z}$ by Theorem A; and $F(D) \in 8\mathbb{Z}$ because D is g-invariant. So (3) gives $\mathrm{Index}(C) + \mathrm{Index}(C_2) \equiv 0 \pmod{48}$, and $\dim C_{2,+} < \dim C_+$. After finitely many steps $C_+ = 0$ and $C = V_-$, a sum of $3k$ self-dual codes of length 4 (tetracodes), which has no maximal codewords. Hence $48 \mid \mathrm{Index}(C)$. □

9. Complements: neighbours and two congruences

Two self-dual codes $C \neq C'$ of length n are *neighbours* if $\dim(C \cap C') = n/2 - 1$ (Kneser [6]).

Lemma 9.1. *Any two self-dual codes of the same length are joined by a chain of neighbours.*

Proof. Let $C_1 \neq C_2$ and $x \in C_2 \setminus C_1$. Then $C' = (C_1 \cap x^\perp) + \langle x \rangle$ is a self-dual neighbour of C_1, and $C' \cap C_2 \supsetneq C_1 \cap C_2$. Induct on $n/2 - \dim(C_1 \cap C_2)$. □

Lemma 9.2 (neighbour identity)**.** *Let $\Lambda \colon \mathbb{F}_3^n \to \mathbb{Z}$ satisfy $\hat{\Lambda} = 3^{n/2}\Lambda$, vanish on anisotropic vectors, and put $S(X) = \sum_{x \in X} \Lambda(x)$. If C, C' are neighbours with $D = C \cap C'$, then $S(C) + S(C') = 4S(D)$.*

Proof. $D^\perp = C + C'$ by dimension. For $c \in C \setminus D$ and $c' \in C' \setminus D$, $c \cdot c' \neq 0$, and the cosets $D + ac + bc'$ with $ab \neq 0$ consist of anisotropic vectors ($x \cdot x = 2ab\, c \cdot c'$). So $S(D^\perp) = S(C) + S(C') - S(D)$, and (P) gives $S(D^\perp) = 3S(D)$. □

Remark 9.3. With $\Lambda = f$ and $12 \mid n$ this gives $\mathrm{Index}(C) + \mathrm{Index}(C') = 4F(D)$, and $F(D)$ is even ($x \mapsto -x$), so $\mathrm{Index}(C') \equiv -\mathrm{Index}(C) \pmod 8$. With Lemma 9.1 and a sum of tetracodes (index 0) as base point, this is a second proof of $8 \mid \mathrm{Index}(C)$.

Proposition 9.4. *Let E be self-dual of length $m \equiv 0 \pmod{12}$ and σ any block system. Then $R_\sigma(E) \equiv \mathrm{Index}(E) \pmod{48}$.*

Proof. Let $\Lambda = w_\sigma$, with s singletons and b blocks. One checks $\hat\lambda = 9\lambda$ on $\mathbb{F}_3^4$ (summing over the words of weight 1 and 4), so by (M) and (T) $\hat\Lambda = (\sqrt{-3})^s 9^b \Lambda = 3^{m/2}\Lambda$, as $s = m - 4b \equiv 0 \pmod 4$. If $\Lambda(x) \neq 0$, then $\operatorname{wt}(x) = s + \sum_\beta w_\beta$ with $w_\beta \in \{1, 4\}$, so $\operatorname{wt}(x) \equiv s + b = m - 3b \equiv 0 \pmod 3$: Λ vanishes on anisotropic vectors. So Lemma 9.2 applies to Λ and to f, and $\Psi = \sum(\Lambda - f)$ satisfies $\Psi(C) + \Psi(C') = 4\Psi(D)$. Now $\Lambda - f$ is even on every vector (on maximal vectors both are ± 1; otherwise $f = 0$ and Λ has a factor $\lambda(1) = -2$ or vanishes), and $x \mapsto -x$ preserves it, so $\Psi(D) \in 4\mathbb{Z}$ and $\Psi(C') \equiv -\Psi(C) \pmod{16}$. For the sum of tetracodes whose blocks include those of σ, $\Lambda = f = 0$. By Lemma 9.1, $R_\sigma(E) \equiv \operatorname{Index}(E) \pmod{16}$; both lie in $24\mathbb{Z}$, by Theorem A for $\operatorname{Index}(E)$ and by Lemma 4.6 and Theorem A for $R_\sigma(E) = \operatorname{Index}(Q_\sigma(E))$. □

So a pair (E, σ) can have $R_\sigma(E) = \pm 24$ only if $\operatorname{Index}(E) \equiv 24 \pmod{48}$; the values $\pm 24, \pm 72$ of $R_\sigma(G_{12})$ in §5 are all $\equiv 24$.

Proposition 9.5. *In every self-dual ternary code the numbers of even and of odd maximal codewords are both divisible by* 8.

Proof. We may assume $12 \mid n$ (Lemma 2.1). Let $U(X)$ be the number of maximal vectors in X and $\Delta = U - F = 2 \cdot \#\{\text{odd}\}$. As in Lemma 9.2, $\Delta(C) + \Delta(C') = \Delta(D^\perp) + \Delta(D)$ and $F(D^\perp) = 3F(D)$. Poisson summation with $\hat\imath$ gives $3^{n/2-1}U(D^\perp) = \sum_{u \in D} 2^{n - \operatorname{wt} u}(-1)^{\operatorname{wt} u}$. Modulo 16 only $\operatorname{wt} u \in \{n, n-3\}$ survive, and the words of weight $n-3$ come in pairs $\pm u$, so $U(D^\perp) \equiv 3^{1-n/2}U(D) \pmod{16}$. As $U(D)$ is even and $3^{1-n/2} \equiv 3 \pmod 8$, $U(D^\perp) + U(D) \equiv 4U(D) \pmod{16}$, and $\Delta(C) + \Delta(C') \equiv 4\Delta(D) = 8\#\{\text{odd in } D\} \equiv 0 \pmod{16}$, since odd vectors of D come in pairs $\pm x$. A sum of tetracodes has $\Delta = 0$, so $16 \mid \Delta(C)$ for all C by Lemma 9.1; hence $8 \mid \#\{\text{odd}\}$, and $8 \mid \#\{\text{even}\}$ by Theorem A. □

10. Computations and formal verification

Formal verification. The repository [9] contains a Lean 4/Mathlib formalization. Its theorem `GJF.conjecture_1_4` states, for codes C given as `Submodule (ZMod 3) (Fin n → ZMod 3)`,

$$\forall n\ \forall C \subseteq \mathbb{F}_3^n : \quad \texttt{IsSelfDual}\, C \ \wedge\ 0 < n \implies 24 \mid \texttt{gjfIndex}\, C, \qquad \text{and}$$
$$\forall k \geq 1\ \exists C \subseteq \mathbb{F}_3^{12k} : \quad \texttt{IsSelfDual}\, C \ \wedge\ \texttt{gjfIndex}\, C = 24,$$

i.e. Theorems A and B, with `IsSelfDual C` meaning $x \in C \iff x \perp C$ and `gjfIndex` the number of even minus the number of odd maximal codewords. It depends only on the axioms `propext`, `Classical.choice`, `Quot.sound`; finite checks (the local identities, the Golay values of Proposition 5.1) are done by the kernel. The divisibility proof follows §3; $|C|^2 = 3^n$ is obtained by Poisson summation of the constant function, so no dimension theory is needed.

Theorem D is formalized as `GJF.Obs.theoremD`, for a signed permutation g with $gC \subseteq C$ and $g^4 = -1$: it gives $4 \mid n$, $48 \mid \operatorname{Index}(C)$, and $\operatorname{Index}(C) = 0$ if $n/4$ is odd. The formal proof follows §8 without the field $\mathbb{F}_9$. The projector onto V_+ along V_- is $g^3 + g - 1$, so $C = (C \cap V_+) \oplus (C \cap V_-)$. For $v \in C \cap V_+$, $v \neq 0$, choose $w \in V_-$ with $v \cdot w = 1$, put $W = \langle w, gw \rangle$ and $D = C \cap W^\perp$. Then $D^\perp = C \oplus W$, and a word $c + y$ ($c \in C \setminus D$, $y \in W$) is isotropic only for $y \in \{0, \pm L(c)\}$, where $L(c) = (c \cdot w)\, gw - (c \cdot gw)\, w$. So the four codes through D are C, $C_2 = D + W$, and the graphs C_3, C_4 of $\pm L$, with $g\, C_3 = C_4$; no classification of Hermitian planes is needed. The induction is on $|C \cap V_+|$ and proves $16 \mid \operatorname{Index}(C)$, using $8 \mid \operatorname{Index}(C_3)$ and $4 \mid F(D)$ (the free action of $\langle g^2 \rangle \cong \mathbb{Z}/4$); Theorem A supplies the factor 3. The repository also formalizes the code-level consequences of Theorem C that the Main Question uses, for all k (`GJF.Ex.tierA`), with an earlier choice of seeds and with the value $R_\sigma(E_{72}) = 24$ as an explicit hypothesis.

Computations. The finite facts used above are:

- Proposition 5.1 and Lemma 5.4: sums over the 729 words of G_{12}.
- $J, K \in \mathrm{Aut}(G_{12})$ and $R_{\sigma_Q}(G_{12}) = -24$: 729 words.
- Proposition 7.3: a word contributing to $R_\sigma(E)$ is nonzero on every singleton, hence determined by its signs on an information set of singletons (and its values on a few block coordinates). This leaves 2^{18} candidates for σ_A and $2^{12} \cdot 3^6$ for σ_B. The twined values are sums over the 3^6 words of E_{36}^h.
- Proposition 7.4: $R_{\sigma_{QZ}}(E_{72})$ is a sum over 2^{36} sign patterns on an information set outside σ_{QZ}. It was computed by three independent programs, one of them via the description of Q_8-invariant codes as $\mathbb{F}_3^2 \otimes U$. The twined value is a sum over the 3^{12} words of E_{72}^h.

In addition, the codes of Theorems B and C were built explicitly to lengths 84 (Theorem B) and 96, 180, 288 (the three families of Theorem C). Self-duality, the group relations and the block types were checked on every code, sign-exact indices at lengths 24, 36, 48, 72, and vanishing twined counts at lengths 72, 108, 144, 180. Scripts and generator matrices are in [9]; one script re-derives every finite fact of this paper from the data printed in it, with its own implementation of Construction 6.3.

Appendix A. Seed data

A.1. **The code E_{36}.** Generator matrix of the code of [4, Ex. 5.7(3)], coordinates $0, \dots, 35$; the automorphism is $h\colon e_i \mapsto e_{i+12}$ (indices mod 36).

```
100000000000000000220222100201111202
010000000000000000120020020002100011 0
001000000000000001000210122010210120
000100000000000001000010211101220210
000010000000000000201221022202011111
000001000000000000202022200202110201
000000100000000001001011021121100001
000000010000000001200022201200120210
000000001000000002102001011211121021
000000000100000002101011110100212212
000000000010000001200201102221112110
000000000001000001001201001120002000
000000000000100000002211111202221012
000000000000010002200211212021221020
000000000000001002000210110101001122
000000000000000102202222102211221211
000000000000000012101012011112210012
000000000000000000012112012121111022
```

A.2. **The code E_{72}.** Identify each block $\{4b, 4b+1, 4b+2, 4b+3\}$ with the quaternion algebra $A = \mathbb{F}_3\langle i, j\rangle$, the four coordinates being those of $a+bi+cj+dk$ ($k = ij$), and A with $M_2(\mathbb{F}_3)$ via $i \mapsto \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $j \mapsto \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. Explicitly, the matrix $\left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$ has coordinates $p(2,0,1,1) + q(0,2,1,2) + r(0,1,1,2) + s(2,0,2,2)$. Let $U \subseteq \mathbb{F}_3^{36}$ be the span of the rows below, the coordinates $2b, 2b+1$ belonging to block b. Then E_{72} consists of the tuples $(X_0, \dots, X_{17})$ of matrices $X_b \in M_2(\mathbb{F}_3)$ such that the first rows of $X_0, \dots, X_{17}$, concatenated, form a vector of U, and so do the second rows. J and K are left multiplication by i and j on every X_b, i.e. on the coordinates of each block $J(e_0, e_1, e_2, e_3) = (e_1, -e_0, e_3, -e_2)$ and $K(e_0, e_1, e_2, e_3) = (e_2, -e_3, -e_0, e_1)$, and h maps block $3s + r$ to block $3s + (r + 1 \bmod 3)$. E_{72} is self-dual because U is Lagrangian for $\sum_b (u_{2b}v_{2b+1} - u_{2b+1}v_{2b})$.

```
100000000000000000122202220220200210
010000000000000000112200100211122110
001000000000000000021222202202102002
000100000000000000001122111002101221
```

```
00001000000000000022021202202021020
00000100000000000022001102111021 1012
00000010000000000022020111212121 1220
00000001000000000020210002101002 2211
00000000100000000001220221112120 2112
00000000010000000000202110021011 0222
00000000001000000002012221211112 2021
00000000000100000021002010100222 1102
00000000000010000021120110001110 1120
00000000000001000011122020012100 0022
00000000000000100001211211100020 1011
00000000000000010020111221200122 0000
00000000000000001012012100111011 2010
00000000000000000112201101212000 2200
```

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